

Quartic Analysis

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We're given

- $\mathbf{X}$: an N cells $\times$ P odours matrix of observed input responses,
- $\mathbf{C}$: a P cells $\times$ P cells observed output covariance matrix,
- The regularization parameter $\lambda = \lambda_0 \frac{P^2}{N}$, where λ_0 is the value we sweep in the paper.
- The mean subtraction operation, $\mathbf{J} = \mathbf{I} - N^{-1}\mathbf{1}\mathbf{1}^T$.
- The prediction error $\mathbf{E}(\mathbf{z}) = \mathbf{X}^T[\mathbf{z}]\mathbf{J}[\mathbf{z}]\mathbf{X} - \mathbf{C}$

Our loss function is

$$L(\mathbf{z}) = \frac{1}{2}\|\mathbf{E}(\mathbf{z})\|_F^2 + \frac{\lambda}{2}\|\mathbf{z} - \mathbf{1}\|_2^2, \quad (1) \quad \{\text{eq:loss}\}$$

where $[\cdot]$ converts its argument to a diagonal matrix.

We want to view this from the perspective of a single unit z_i . To do that, we rewrite $\mathbf{E}(\mathbf{z})$ to isolate that unit. Examining its components,

$$\begin{aligned} \mathbf{X}^T[\mathbf{z}]\mathbf{J}[\mathbf{z}]\mathbf{X} &= \mathbf{X}^T[\mathbf{z}^2]\mathbf{X} - N^{-1}\mathbf{X}^T\mathbf{z}\mathbf{z}^T\mathbf{X} \\ \mathbf{X}^T[\mathbf{z}^2]\mathbf{X} &= \sum_{j=1}^N \mathbf{x}_j \mathbf{x}_j^T z_j^2 = \mathbf{x}_i \mathbf{x}_i^T z_i^2 + \sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j^T z_j^2 \\ \mathbf{m} &\triangleq \mathbf{X}^T \mathbf{z} = \sum_j \mathbf{x}_j z_j = \mathbf{x}_i z_i + \sum_{j \neq i} \mathbf{x}_j z_j = \mathbf{x}_i z_i + \mathbf{m}_i \\ \mathbf{E}(\mathbf{z}) &= \mathbf{x}_i \mathbf{x}_i^T z_i^2 + \sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j^T z_j^2 - N^{-1}(\mathbf{x}_i z_i + \mathbf{m}_i)(\mathbf{x}_i z_i + \mathbf{m}_i)^T - \mathbf{C} \\ &= \mathbf{x}_i \mathbf{x}_i^T z_i^2 - N^{-1}(\mathbf{x}_i \mathbf{m}_i^T z_i + \mathbf{m}_i \mathbf{x}_i^T z_i + \mathbf{x}_i \mathbf{x}_i^T z_i^2) + \sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j^T z_j^2 - N^{-1} \mathbf{m}_i \mathbf{m}_i^T - \mathbf{C} \\ &= \alpha \mathbf{x}_i \mathbf{x}_i^T z_i^2 - N^{-1}(\mathbf{x}_i \mathbf{m}_i^T + \mathbf{m}_i \mathbf{x}_i^T) z_i + \mathbf{E}_i. \end{aligned}$$

Computing the goodness-of-fit term of the loss,

$$\begin{aligned} \text{tr}(\mathbf{E}(\mathbf{z})^T \mathbf{E}(\mathbf{z})) &= \alpha^2 \|\mathbf{x}_i\|_2^4 z_i^4 - 4\alpha N^{-1} \|\mathbf{x}_i\|_2^2 \mathbf{m}_i^T \mathbf{x}_i z_i^3 + 2\alpha \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i z_i^2 \\ &\quad + 2N^{-2} z_i^2 (\|\mathbf{x}_i\|_2^2 \|\mathbf{m}_i\|_2^2 + (\mathbf{x}_i^T \mathbf{m}_i)^2) - 4N^{-1} z_i \mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i + \text{tr}(\mathbf{E}_i^T \mathbf{E}_i). \end{aligned}$$

Our loss function in terms of z_i is then

$$\begin{aligned}
L(z_i) = & \frac{1}{2}\alpha^2\|\mathbf{x}_i\|_2^4 z_i^4 \\
& - \frac{2\alpha}{N}\|\mathbf{x}_i\|_2^2 \mathbf{m}_i^T \mathbf{x}_i z_i^3 \\
& + \left(\alpha \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i + \frac{\|\mathbf{x}_i\|_2^2 \|\mathbf{m}_i\|_2^2 + (\mathbf{x}_i^T \mathbf{m}_i)^2}{N^2} + \frac{\lambda}{2} \right) z_i^2 \\
& - \frac{2\mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i + N\lambda}{N} z_i \\
& + \frac{\text{tr}(\mathbf{E}_i^T \mathbf{E}_i) + \lambda(1 + \sum_{i \neq j} (z_j - 1)^2)}{2}.
\end{aligned}$$

The trailing constant doesn't matter for the minimization, so we'll drop it. We'll express the remaining coefficients in terms of a few geometric quantities:

- The raw power of the population mean $m \triangleq \mathbf{m}_i^T \mathbf{m}_i$;
- The raw channel power, $p_i \triangleq \mathbf{x}_i^T \mathbf{x}_i$;
- The residual channel power, $q_i \triangleq \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i$;
- The raw alignment with the population mean, $a_i \triangleq \mathbf{x}_i^T \mathbf{m}_i$;
- The residual alignment with the population mean, $b_i \triangleq \mathbf{x}_i^T \mathbf{E}_i \mathbf{m}_i$;
- The curvature at the origin, $\kappa_i \triangleq \alpha q_i + \frac{p_i m + a_i^2}{N^2}$.

In these terms, and dropping the constant,

$$L(z_i) = \frac{1}{2}\alpha^2 p_i^2 z_i^4 - \frac{2\alpha}{N} p_i a_i z_i^3 + \left(\kappa_i + \frac{\lambda}{2} \right) z_i^2 - \left(\frac{2b_i}{N} + \lambda \right) z_i \quad (2) \quad \{\text{eq:Lzi}\}$$

$$\triangleq A z_i^4 + B z_i^3 + C z_i^2 + D z_i. \quad (3) \quad \{\text{eq:Lzi}\}$$

To better understand this quartic, we will change coordinates. First, we'll rescale by A ,

$$\begin{aligned}
L(z) & \propto z^4 + \frac{B}{A} z^3 + \frac{C}{A} z^2 + \frac{D}{A} z \\
& \triangleq z^4 + b z^3 + c z^2 + d z.
\end{aligned}$$

Next, we'll shift to a coordinate $s = z - \bar{z}$ in which the cubic disappears. Expressing the loss as a function $\tilde{L}$ of s , we note that for a monic quartic with no cubic, the third derivative should be $24s$.

$$\begin{aligned}
\tilde{L}(s) & = (s + \bar{z})^4 + b(s + \bar{z})^3 + c(s + \bar{z})^2 + d(s + \bar{z}). \\
\tilde{L}'''(s) & = 24(s + \bar{z}) + 6b = 24s \implies \bar{z} = -\frac{b}{4}.
\end{aligned}$$

So, in the shifted and scaled coordinates, the loss function is

$$\begin{aligned}
\tilde{L}(s) & = \left(s - \frac{b}{4} \right)^4 + c \left(s - \frac{b}{4} \right)^2 + d \left(s - \frac{b}{4} \right) \\
& = s^4 + \tilde{b} s^2 + \tilde{c} s + \tilde{d}.
\end{aligned}$$

We can expand out the first equation to get the coefficients. But an easier way is to Taylor expand our loss function $L(z)$ to get a polynomial in s :

$$\tilde{L}(s) = L(s + \bar{z}) = \underbrace{L(\bar{z})}_d + \underbrace{L'(\bar{z})}_c s + \underbrace{\frac{1}{2}L''(\bar{z})}_b s^2 + \frac{1}{24}L'''(\bar{z})s^4.$$

Then, $L''''(\bar{z}) = 24$. The other derivatives are

$$L'(\bar{z}) = 4\bar{z}^3 + 3b\bar{z}^2 + 2c\bar{z} + d = -\frac{b^3}{16} + 3\frac{b^3}{16} - \frac{bc}{2} + d = \frac{b^3}{8} - \frac{bc}{2} + d$$

$$L''(\bar{z}) = 12\bar{z}^2 + 6b\bar{z} + 2c = 12\frac{b^2}{16} - 6\frac{b^2}{4} + 2c = -\frac{3}{4}b^2 + 2c.$$

So we arrive at

$$\tilde{L}(s) = s^4 + \left(c - \frac{3}{8}b^2\right)s^2 + \left(d - \frac{bc}{2} - \frac{b^3}{8}\right)s + \text{const.}$$